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The Origins of Perceived Discrimination in e-Learning in Nursing Students: A Qualitative Study

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Abstract

Introduction: Discrimination in education is commonly associated with face-to-face interactions between teachers and students. However, e-learning environments can also foster discrimination. Despite existing research on discrimination in traditional education, limited studies address this issue in virtual education, particularly in nursing education.

Objectives: This study aimed to explore the factors contributing to perceived discrimination in e-learning among nursing and midwifery students, focusing on the underlying causes and conditions that shape these experiences.

Methods: A qualitative design was employed at the nursing and midwifery school of Isfahan University of Medical Sciences. Thirteen nursing and midwifery students were selected using purposeful and convenient sampling. Data were collected through semi-structured, in-depth interviews, and analyzed using content analysis.

Results: Factors contributing to perceived discrimination in e-learning were categorized into four themes: (1) the nature of e-learning (e.g., limited teacher availability, reduced interaction); (2) professor-related factors (e.g., experience, age, technology skills, and inattention to feedback); (3) student-related factors (e.g., gender, financial issues, and varying technology skills); and (4) inadequate resources (e.g., internet connectivity and limited access to study materials).

Conclusions: This study highlights multiple factors influencing students' perceptions of discrimination in e-learning. Addressing these issues can improve virtual education quality in nursing programs. Further research is needed to explore these factors in broader educational contexts

Keywords

discrimination, injustice, virtual education, nursing students, qualitative research

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Introduction

e-Learning is an experience that one third of university students have encountered (McNeill et al., 2016). As the number of students applying for higher education rises, the availability of online classes and the overall volume of online coursework are also increasing (Makda, 2024). Generally, this educational method represents one of the most significant applications of information technology (Lee et al., 2024), particularly following the coronavirus pandemic (Abbasnejad et al., 2024). Since the advent of e-learning, the use of computers in nursing education has expanded considerably (Haanes et al., 2024). Furthermore, e-learning has been referenced in numerous nursing articles (McDonald et al., 2018).

e-Learning offers several advantages, including the flexibility to learn from any location and at any time, the ability to revisit lessons, a shift from a teacher-centered approach to a student-centered model, and a reduction in educational costs (Hosseini et al., 2016). However, this form of education also faces criticism. For instance, Ifeus, a contemporary professor of philosophy and a critic of the Internet, argues that

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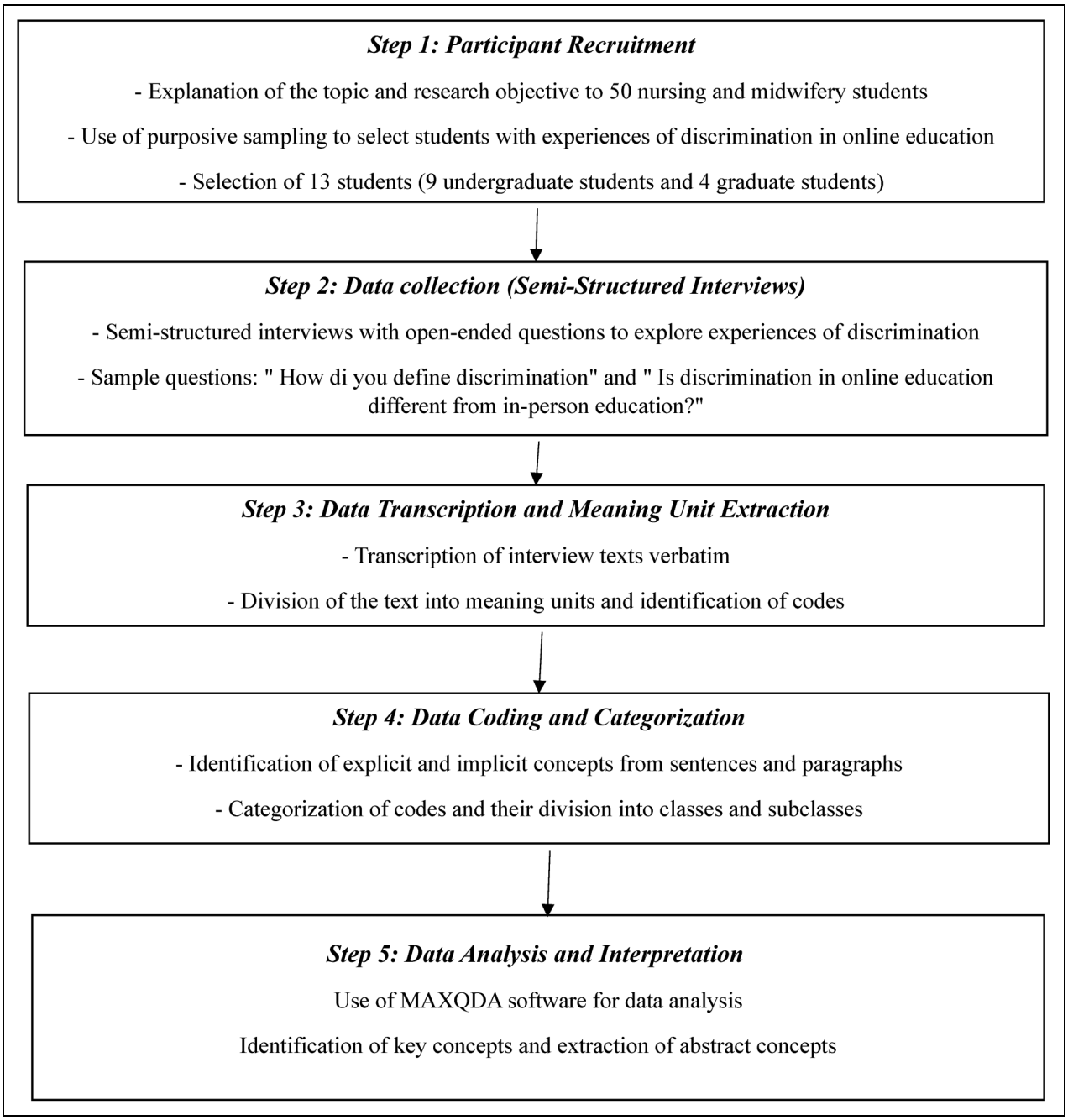

Figure 1. Data collection and analysis process.

e-learning cannot guarantee the development of creative ideas, the quality of information, users' comprehensive understanding of various realities, or the potential for a meaningful life for learners (Dreyfus, 2010). Among the disadvantages of e-learning highlighted in the articles are unequal access to educational resources, slow internet connectivity in certain regions, high costs of internet services, and a lack of information regarding the use of technology and facilities in rural areas (Bello et al., 2017). Barker et al. (2013) identified the implementation of e-learning for students living in impoverished and rural areas as a significant challenge (Barker et al., 2013). In a study on e-learning, non-native students reported that, unlike their native counterparts, professors paid little attention to them or even ignored them during lectures due to their unfamiliarity with the professors (Ilonga et al., 2020). These factors may lead students to feel unfairly treated or discriminated against in e-learning environments. It is also important to note that online educational platforms can potentially serve as a breeding ground for discrimination due to cultural differences (Zembylas, 2008).

Discrimination is defined as a biased and harmful distinction (Fadakar Davarani et al., 2018), while injustice is described as a disparity between what is and what ought to be (Gilbert et al., 2013). To implement effective education, establishing equity is crucial (Gilbert et al., 2013). Educational equity is defined as the perceptions of outcomes or processes that occur within the educational context (Berti et al., 2010). It refers to providing equal educational opportunities for all students, which often involves ensuring fairness in professors' interactions with students, particularly in classes characterized by significant diversity and differences among learners (Mosher, 2010). This form of justice encompasses interactions, behaviors, and performance methods grounded in fairness, impartial treatment, and guidance tailored to students' abilities. It also emphasizes equitable assessment and grading practices, which can foster a sense of self-worth among students, enhance their self-confidence, and cultivate a positive academic attitude. Ultimately, this approach contributes to the development of desirable civic and academic behaviors (Marzoghi et al., 2013). In contrast, experiences of discrimination are likely to indirectly influence students' developmental competencies and are currently recognized as significant stressors (Benner & Graham, 2011). Research has documented the detrimental effects of discrimination on both mental and physical health, leading to symptoms such as depression, high blood pressure, and increased cardiovascular risks (Chun et al., 2015). For instance, discrimination within the Indian education system has adversely affected not only the educational and academic status of students but also their psychological well-being (Sharma & Subramanyam, 2020).

Review of Literature

Few studies have examined discrimination, and they have primarily focused on face-to-face education. According to Wang and Huguley (2012), discrimination in teaching and learning can negatively impact academic achievement indicators, including semester grade point average (Wang & Huguley, 2012). In another study, African American students reported that teachers treated them with less respect and that they felt less intelligent than their peers, which contributed to lower self-esteem (Brittian & Gray, 2014). Similarly, a study conducted by Hadian Jazi et al. (2022) found that feelings of discrimination led to a decrease in self-confidence among nursing students. Consequently, nursing instructors and professors should seek solutions to prevent the emergence of such feelings by refraining from inappropriate behavior and discriminatory language (Hadian Jazi et al., 2022). These issues are critically important for nursing students, as they can lead to psychological problems and diminished self-confidence, ultimately jeopardizing patient safety in clinical settings (McNeill et al., 2016). Furthermore, these challenges may result in unethical behavior among nurses and nursing students, such as inaccurately recording vital signs or altering patient prescriptions (Clark, 2006).

According to the studies mentioned above, research on discrimination has primarily focused on face-to-face education. Given the negative impact of discrimination in these educational settings on students—particularly nursing students, due to concerns regarding patient safety—and the rise of e-learning, especially following the coronavirus pandemic, it is essential to consider can discrimination have detrimental effects in e-learning environments for nursing student? Therefore, the aim of the present study was to investigate nursing students' experiences of discrimination in e-learning through a qualitative approach. Qualitative research, in contrast to quantitative research, offers researchers greater opportunities to uncover and elucidate the complexities of the educational environment. It enables a deeper understanding of various challenging aspects and provides valuable insights into the realities faced in the field (Majid & Vanstone, 2018).

Methods

Study Design

This study was conducted using a qualitative research approach, specifically conventional content analysis as described by Hsieh and Shannon (2005). This method was chosen due to its suitability for exploring phenomena with limited prior knowledge. Conventional content analysis allows categories and themes to emerge directly from the data, enabling a deeper understanding of participants' experiences without the influence of preconceived frameworks (Hsieh & Shannon, 2005). This approach was deemed appropriate for examining the phenomenon of perceived discrimination in e-learning, as it facilitated the identification of both explicit and implicit patterns within the data. The study was underpinned by a content analysis methodological orientation, ensuring a systematic exploration of participant experiences.

Setting-Participants

This study was conducted in 2022–2023 at the School of Nursing and Midwifery of Isfahan University of Medical Sciences. The study population included bachelor's, master's, and doctoral students. However, during the sampling process, only bachelor's and master's students who met the inclusion criteria were willing to participate. Purposeful sampling was used to maximize the depth and richness of data (Speziale et al., 2011). The researchers first explained the topic and purpose of the research to approximately 50 nursing and midwifery students by phone or in person. Students who expressed their readiness to participate were screened based on the following inclusion criteria: (1) being a nursing or midwifery student at the Isfahan University of Medical Sciences, (2) having completed at least one virtual course, and (3)

having experienced perceived discrimination during e-learning. Students who did not meet these criteria or declined to participate were excluded. Ultimately, 13 students (nine bachelor's and four master's students) participated in the study. The researchers included both male and female participants to ensure diversity in perspectives. Participants were approached face-to-face or by phone, and no dropouts occurred during the study.

Data Collection

The principal approach employed for data gathering in this research comprised semi-structured open-ended interviews. In these sessions, participants were prompted to share their encounters pertaining to discrimination within the domain of online education. Sample questions posed to participants encompassed inquiries such as: How would you define discrimination? Does discrimination within the virtual educational setting differ from discrimination encountered in traditional face-to-face instruction? Have you, at any point, experienced feelings of discrimination from instructors during virtual instruction? If so, what elements contributed to this sentiment? Furthermore, what repercussions has this perceived discrimination had on the quality of your educational experience and learning outcomes? Follow-up questions were asked based on the information provided by the participants to clarify the concept under study. To encourage participants and to gain more in-depth information, exploratory questions such as "be further explained?" Or "give an example" were asked. The interview guide was pilot-tested prior to the main study to ensure clarity and relevance of the questions.

All interviews were conducted by the first author, who is a PhD holder and university assistant professor with extensive experience in qualitative research and nursing education. The interviewer was female and had prior training in qualitative interviewing techniques. A relationship with participants was not established prior to the study; however, participants were informed about the researcher's academic background, goals, and interest in the topic. Interviews were audio-recorded and supplemented by field notes to capture non-verbal cues and contextual details. Each interview lasted approximately 45–60 min. Data saturation was achieved after interviewing 13 participants, with no new themes emerging in the final interviews.

Data Analysis

At the end of each interview, data were analyzed by content analysis based on the five steps of Graneheim and Lundman by researchers (Graneheim & Lundman, 2004). In the first step, the text of the interviews was immediately implemented verbatim on the recording tape and used as the main research data (Transcribing). In the second step, the recorded audiotape listened to several times, the handwritten texts were reviewed several times and a decision was made to divide the text into meaning units. In the third step, the abstracting design of semantic units and selection of codes was performed. According to the participants' experiences, overt and covert concepts were identified in the form of sentences or paragraphs of their words and signifying codes. Then coding and summarizing were done. In the fourth step, based on the continuous comparison of similarities, differences, and the appropriateness of the codes that indicate a single subject, were placed in a class and were subdivided into classes and sub-classes, and axial codes were formed. Lastly, at the interpretive level, the summary classes and the central concept of each class were identified and the main and abstract concepts were extracted. Then data analysis was performed in the MAXQDA version 10.

Two researchers independently coded the data to ensure reliability. A coding tree was developed to organize and categorize codes systematically, and themes were derived inductively from the data. Participants were invited to review and confirm the accuracy of the transcripts and the interpretation of findings to enhance the credibility of the study.

In this study, data were saturated after 13 students. Saturation in qualitative studies means that the researchers seek to replicate and confirm the previously collected data. Lincoln and Guba's criteria were used to determine the validity of the data. To confirm the data, the researcher's long-term involvement, appropriate interaction with the participants, and their review were used, which also helped to increase the data acceptability. Also, the combination of time and variety of sampling increases the validity of the data and leads to the verifiability of the data (Figure 1).

Table 1. Demographic and Relevant Characteristics of Participants.

Variables	Participants ($N = 13$)
Age ($M \pm SD$)	23.76 ± 0.69
Gender (N, %)	Male: 4 (31%), Female: 9 (69%)
Grade (N, %)	Bachelor: 7 (53%), Master: 6 (47%)
Previous e-learning experience (N, %)	Yes: 8 (61%), No: 5 (39%)
Digital literacy (N, %)	High: 5 (38%), Medium: 6 (46%), Low: 2 (16%)

Results

In this study, 13 nursing students from bachelor's and master's programs participated. The demographic and relevant characteristics of the participants are presented in Table 1. This table includes not only the basic demographic information (age, gender, and grade) but also additional characteristics such as previous e-learning experience and digital literacy, which were considered important for understanding the participants' perspectives on e-learning.

The factors causing the feeling and experience of discrimination in e-learning were categorized into four major themes,

Table 2. Summary of Extracted Themes, Categories, and Subcategories Related to Perceived Discrimination in e-Learning.

Theme	Category/Subcategory
The nature of e-learning	1. Less teacher availability 2. Reduced interaction between professor and students
Factors related to the professor	1. Professor's experience 2. Professor's age 3. Professor's field of work 4. Professor's skill in using technology 5. Inattention to criticisms
Factors related to students	1. Gender 2. Field of study 3. Financial problems 4. Differences in ability to use technology
Inadequate resources	1. Internet connection issues 2. Lack of access to study and library resources

which are presented in Table 2. This table summarizes the extracted themes, categories, and subcategories related to perceived discrimination in e-learning. The themes reflect the multifaceted nature of discrimination as experienced by nursing students in e-learning environments, with a focus on the nature of e-learning, factors related to professors, factors related to students, and inadequate resources.

To provide a clearer understanding of how the final themes were derived from the interview data, Table 3 presents the process of theme derivation. This table illustrates how initial codes (about 180) derived from participant quotes were grouped into categories and further abstracted into broader themes. It includes specific examples of participant quotes and how these were linked to the final themes. This process highlights the step-by-step analysis conducted during the research, ensuring transparency in the coding and theme development process.

The Nature of e-Learning

The factors encompassed within this category arise from the intrinsic characteristics of e-learning and are essentially unavoidable. Indeed, a significant proportion of the elements that contribute to students' perception of discrimination in virtual education stem from the limited interaction between the professor and the student—a consequence inherent to the nature of this educational modality.

Comparatively, in traditional face-to-face education, the presence of direct communication and the utilization of nonverbal cues result in fewer instances of misinterpretation regarding the professor's demeanor and discourse. However, in virtual education, where only the professor's voice is audible, the potential for misunderstandings leading to a student's mistaken or valid sense of discrimination is heightened.

Regarding the reduced student–teacher interaction, one participant stated:

> In e-learning, there was no or very little interaction between teacher and student; for example, in the face-to-face class, the teacher noticed from the faces that who understood the lesson, in virtual training however, the absence of facial cues makes it challenging for the teacher to discern who comprehends the material. Additionally, students who actively participate and have their names displayed on the screen would naturally draw the teacher's attention. Some students even employed a strategy of speaking or typing more as a means to capture the professor's attention. However, it became evident that their increased engagement did not necessarily correlate with their attentiveness to the lesson content. This situation led me to feel that favoritism was being shown towards these students, resulting in a sense of discrimination. (Participant 2)

Another participant stated:

> In e-learning, when one person speaks, it is no longer possible to exchange opinions with others at the same time (unlike face-to-face training) and others may feel discriminated against. Indeed, within the area of virtual education, opportunities for students to engage in mutual discussions and express their viewpoints are infrequent or restricted. I've observed instances where the professor grants one student the chance to speak, and just as others prepare to share their thoughts, the professor declares time constraints, preventing further comments. This scenario triggers emotions of confusion and bias, leading me to question why my peer was allowed to speak while I was cut off due to time limitations. This experience fosters a sense of discrimination, even though I recognize that this sentiment might not be entirely accurate. In the heat of the moment, however, these feelings of being unfairly treated become very real. (Participant 5)

Another element inherent to the nature of e-learning that can potentially foster discrimination among students is the unequal access to instructors. This concern is particularly pronounced for students who reside in a different city than their academic institution. Conversely, those students who reside within the same city as their educational establishment enjoy a more convenient avenue for reaching out to and engaging with their professors. This distinction often results in a disproportionate allocation of attention and importance from the professor towards local students, irrespective of the caliber of assignments or the progression within the course. Consequently, students not situated in the immediate vicinity might experience feelings of being unfairly treated and marginalized. Examples of participants' comments are as follows:

> Participant 9: Local students who already knew the professor prepare special files and hand them over to the professor to increase the grade, and are more in touch with the professor, but students who cannot do so due to distance feel discriminated.

Table 3. Process of Deriving Final Themes from Interview Data With Examples of Participant Quotes.

Participant Quote	Initial Code	Category/Subcategory	Final Theme
"I feel like the professor doesn't have enough time to answer my questions."	Lack of professor availability	Factors related to the professor (availability)	Nature of e-learning
"I can't interact with my classmates as much as I would in a face-to-face class."	Reduced interaction with classmates	Nature of e-learning (interaction)	Nature of e-learning
"My professor is not familiar with using the online platform very well."	Lack of technical skills	Factors related to the professor (technology skills)	Factors related to the professor

> Participant 3: Usually during the e-learning, we did not have the teacher's phone number, or because most of the time the teacher did not see us in person and did not recognize our numbers, he did not answer; or we felt we were disturbing the teacher. And if we asked our questions by texting or email, most of us would not have received an answer and only students whom the professor already knew or who were smarter students would've received an answer. And we felt discriminated against.

Factors Related to the Professor

Factors that fell into this category included the professor's experience and age, professor's field of work, their skill in using technology, and inattention to criticisms which created a sense of discrimination among students.

Many students stated that older and more experienced professors made more differences among students than younger and less experienced professors. One of the students said: "Maybe it's defendant on the age of the professors, for example, older professors leave some responsibilities to one person, but the young professors are more thoughtful of not causing any discrimination" (Participant 2).

Truly, it remains challenging to justify why a correlation exists between a professor's age and experience and the heightened perception of discrimination among students in their interactions. Nonetheless, a plausible explanation might be that younger and less experienced educators tend to be more inclined to provide equal educational opportunities to all students, as opposed to favoring specific individuals. This could be attributed to their greater vigor, enthusiasm, and motivation. Their closer age proximity to students might facilitate better understanding, fostering a more comfortable environment for interaction. While this notion remains speculative, it raises a hypothesis that necessitates further exploration through comprehensive studies within this domain.

Not only the age and experience of the professor, but also the type of their academic background can be another factor that causes discrimination among students. This influence becomes especially notable in virtual classrooms encompassing broad or less specialized subjects, drawing participation from students representing diverse disciplines. During instruction, instructors might lean towards examples and illustrations aligned with their principal area of expertise, inadvertently excluding students from other fields. This inclination can engender feelings of marginalization and bias among those students who hail from different academic domains. In confirmation of this statement, one of the nursing students said:

> Some of the courses that were shared with medical students, most of the examples the professors make were related to medical courses, or he/she clearly paid more attention to students whose field was medicine; Sometimes it even happens that when we wanted to ask a question in the virtual class and turned on the question icon, the professor first asked us what field you are in. And if we said nursing, we would feel exactly that the tone of the voice or the professor's answer to the question would change, and the nursing students would feel discriminated against the medical students. (Participant 11)

Therefore, it seems that if the professor teaches in courses where the audience is different students from different fields, he/she should try to use diverse examples related to all fields or give an example that is not related to any particular field in the class.

The teacher's skill in using technology was also a factor that was repeatedly mentioned by the students as a factor in creating a sense of discrimination. In this regard, Participant 8 stated:

> Some teachers did not know how to work with technology, for example, when students held the hands icon (means I have a question), they did not understand at all, or for example, they told students "No one had a question!" and did not look at the chatbox at all.

Or another student said:

> When the professor was teaching, he asked a question and asked us to answer. I was almost the first one to turn on the microphone icon and I was waiting for the teacher to give me access so that I could answer; But the professor didn't know how to create this access for the student and a few minutes passed before he was finally able to give access to the student and at this time there were other students who had given permission to access the microphone and the professor gave them permission. In case it was me who had to answer as the first person and then I felt that the teacher did not respect my right. (Participant 6)

Hence, it is apparent that virtual education and instruction within a computer-mediated environment demand specific knowledge and competencies. In the absence of this expertise, educators might inadvertently engage in practices that inadvertently generate behaviors capable of triggering feelings of discrimination among students.

Another issue that was widely pointed by participants as discriminatory was the indifference to students' criticism by the professors. In general, students believed that e-learning responds much less to students' criticisms than face-to-face education, or that professors do not pursue problems at all.

> Participant 3: In face-to-face meetings, students may tell the professor that you are discriminating, but in virtual training, students are less likely to do so because they know the teacher does not care or follow-up at all.

Another student said:

> In e-learning, criticism is futile, or the teacher does not care at all. Even if we send a criticism by e-mail. In general, things may happen in virtual education that is so imperceptible that one does not know whether there is discrimination or not. (Participant 6)

Consequently, students who hold reservations about the professor's performance within the virtual education context or hold concerns regarding their grades often refrain from voicing their critique due to the absence of in-person access. Despite withholding their objections, a persistent sentiment endures, where they believe that students in traditional face-to-face instruction possess a distinct advantage. These students can directly communicate their observations regarding the professor's teaching efficacy or express criticism, reinforcing the perception of disparity between the two instructional modes.

Student-Related Factors

In this category, factors such as student gender, field of study, economic problems, and differences in ability to use technology were factors that caused a sense of discrimination among students.

The gender of students is identified as a discernible factor contributing to perceived discrimination in both traditional face-to-face and virtual educational settings. The study highlights that a considerable number of female students expressed a sentiment that male professors exhibit a greater inclination toward male students, and conversely, male students perceived that female professors exhibit a stronger focus on female students.

A participant stated:

> One of our professors, who was a man, pays more attention to the boys in general. One of the male students might have asked a ridiculous question and the professor would have answered it warmly, but I (the girl) might have asked an unnecessary question, but he treated me badly, and he said: why do you ask this question, and you did not study well. (Participant 4)

In several instances, it appears that gender-based discrimination occurs inadvertently and without conscious intent on the part of professors. Consequently, it becomes imperative for instructors to remain vigilant about such potential perceptions of discrimination and to make conscientious efforts in affording equitable opportunities to both male and female students, thereby mitigating the emergence of such circumstances.

Students' financial problems were also a factor that prevented some students from having a good laptop or smartphone and being able to attend virtual classes on time or use a microphone or chat room during classes. So, regardless of the student's potential, he or she would miss the opportunity to participate more in the classroom and therefore be discriminated against more than the students who had better facilities.

Participant 3 said:

> There were students for whom even access to the Navid system (a software for presenting lessons virtually) was difficult, and many did not have a good mobile phone or Laptop or did not have Internet access, and it is considered discrimination.

Or Participant 7 said:

> From the very beginning of the Covid-19 pandemic, when virtual education was introduced, some schools, such as dentistry and medicine, were given more facilities and equipment to hold online and offline classes than the school of nursing! For example, e-learning systems were provided for them earlier, and nursing students felt discriminated against compared to medical students.

Further experiences were also mentioned by the students:

> The difference in students' ability to use technology was another factor. Students that could type faster were able to give the answers more quickly and the ones who typed slower felt discriminated against. (Participant 2)

> There was a student who encountered challenges in utilizing a microphone or lacked familiarity with navigating the virtual learning platform to pose questions. Consequently, this situation engendered a sense of disadvantage, further exacerbating the discrepancy between individuals possessing superior computer proficiency yet potentially lesser academic aptitude. (Participant 6)

The factor of "knowledge and skill of using technology" remains a significant contributor to the sensation of discrimination experienced by both professors and students. Therefore,

it becomes highly crucial that, prior to the implementation of any virtual course, adequate assurance is provided regarding the availability of essential prerequisites for this educational format. Equally important is the proficiency of both the professor and the student in navigating technology, warranting careful consideration.

Inadequate Resources

Inadequate e-learning resources in Iran were another factor that seemed to be widely and universally reducing the quality of this kind of education, and ultimately creating a kind of injustice in this type of education and a sense of discrimination among students.

One student said,

> Interruption in the voice is another issue. In some areas the internet connection was very poor. The inability to afford the internet price was also a problem facing many individuals. Students may miss some classes due to poor infrastructure and lack of internet access. Therefore, students from deprived and remote areas felt discriminated against. (Participant 1)

Another factor that falls into this category was the lack of access to adequate teaching and library resources during the COVID-19 pandemic and e-learning eras. Participant 9: "when students do not have equal access to library resources, especially for non-locals, they feel a discrimination." Or Participant 11: "The problem with the distance was that people living in other cities did not have access to reference books or had very limited access to the Internet."

Unfortunately, in numerous countries, Iran among them, a notable bias is evident favoring major urban centers and premier educational institutions. This preference is evident across various domains, encompassing resources like internet accessibility, among others. Notably, the divide widens as one moves farther from the capital city, resulting in a discernible deficit in amenities. This stark contrast invariably fosters sentiments of discrimination among residents and students hailing from these less privileged regions, particularly when compared to their counterparts in larger cities and more esteemed universities. This pressing concern warrants the earnest attention of relevant stakeholders, necessitating concerted efforts to address and redress this issue effectively.

Discussion

This study aimed to investigate nursing students' experiences of feeling discriminated against in virtual education. In this study, students' views on their experience of discrimination from e-learning are divided into four categories: "The nature of e-learning," "professor-related factors," "student-related factors," and "inadequate resources" were classified.

One of the factors mentioned in this study as the cause of discrimination in students was the nature of e-learning. Students believed that in e-learning, the possibility of getting answers to their questions, and exchanging opinions about them was very low, and also due to the lack of face-to-face interaction, students who chatted more were more at the center of the professors' attention. In a study by Telford and Senior (2017), nursing students stated that feedback is needed to ensure effective training, which is very rare in e-learning, and for better effectiveness, e-learning should be combined with face-to-face training (Telford & Senior, 2017) In this regard, in another study, nursing students stated that virtual education prevents them from communicating with friends, classmates, and professors, and this affects negatively learning (Bdair, 2021) or Lakbala (2016) also mentioned the lack of proper communication between students and professors as one of the main challenges of virtual education and the rate was 68% (Lakbala, 2016). Students believed that most e-learning courses were limited to offline ones, without direct contact with the instructor. The students mentioned that the professors only talk about the content, without any other connection such as greeting or saying goodbye, or giving examples to better understand the topic. They noted that using a uniform tone of voice while teaching without observing the teacher's body language reduces the effectiveness of communication. In addition, students stated that not receiving or expecting feedback from a teacher on assignments, disrupts the communication process and reduces their desire to send more messages to the teacher (Salmani et al., 2022) In response to the cases mentioned, based on the study of Harerimana and Mtshali (2018), nursing professors stated that they spend too much time preparing virtual content. They do not even have the time to take care of the needs of their families, and when such work pressures increase, the quality of teaching and interaction with students are among the first victims (Harerimana & Mtshali, 2018). In the study of Keshavarzi et al. (2019), according to one of the faculty members, this type of training is very time-consuming, for example, preparing the content for a 2-hr class, takes 6 hr, and according to another, despite spending time, many were unable to prepare a training file due to the large file size, and it was not possible to convert the file into smaller parts, and this factor caused all the efforts to be lost. Therefore, such issues lead to lack of sufficient time to hold online classes as well as uploading poor-quality content (Keshavarzi et al., 2019). Professors in virtual education should be mindful of their behavior and ensure that they do not favor native students over non-native students. It is essential to treat all students equally without differentiation. Additionally, professors should strive to conduct their classes online rather than offline. If their proficiency with electronic equipment is lacking, it is advisable for university and college officials to organize specialized training sessions to enhance their skills.

In this study, it was mentioned that non-local students, unlike local students, did not interact with professors, which caused a sense of discrimination in them. In a study by Ilonga et al. (2020), students stated that sometimes the grades were entered incorrectly in the system for them, and due to the long-distance, unlike the locals, they cannot get their score in

person from the teacher and follow up (Ilonga et al., 2020). Professors should recognize that while native students can physically advocate for their rights and grades, they must also create similar opportunities for non-native students. For instance, they could organize virtual meetings at designated times, allowing non-native students to express their opinions and concerns.

In the present study, one of the issues raised by students as a cause of discrimination was their gender. In the late 1980s, a meta-analysis was conducted with more than 80 studies on the gender differences that teachers make among students, noting that in mixed classes, male students received more teacher attention than female students. And perhaps because boys have more interaction with the teacher than girls (Beaman et al., 2006; Yilmaz, 2012). So, professors on virtual platforms should pay closer attention to this issue, considering the growing sensitivity of students to gender discrimination. They should strive to include an equal number of both genders in class discussions and provide them with equal attention.

Another issue that exacerbated the feeling of discrimination was the type of student major of study, and it was stated that professors in shared virtual classrooms pay more attention to medical students. In general, the feeling of discrimination against medical students is an issue that most nursing students experience, and in Iran, it is a long-standing issue (Hadian Jazi et al., 2022). Regarding this theme, in the study of Nakhaee and Nasiri (2017), nurses stated that they are not involved in clinical decisions, which caused a feeling of being ignored (Nakhaee & Nasiri, 2017). Therefore, professors in e-learning should give equal attention to all students, regardless of their field of study. Focusing too much on students in a particular discipline can lead to boredom, a loss of focus, and disengagement among students in other fields.

Another issue that students cited as a cause of discrimination was economic problems. They noted that students who were unable to purchase or access technology appropriate to their e-learning education felt discriminated against by their peers. In the study of Bello et al. (2017), 68.5% of students complained about expensive internet shopping (Bello et al., 2017) and in another study, students mentioned that to overcome the problems related to internet speed, they had to buy high-speed internet and the costs associated with this case are very high (Ilonga et al., 2020). Also, in the study of Bdair (2021), students stated that because there is more than one student in their family, providing computers and e-learning equipment for everyone is very expensive and beyond the financial means of families (Bdair, 2021). In the study of Barker et al. (2013), inequality was raised as an important issue because all students did not have fair access to computers, and this inequality was more pronounced in poorer and more prosperous geographical areas (Barker et al., 2013). It is recommended that students who lack the financial means to purchase the necessary electronic devices be identified and referred to the appropriate support centers for assistance before deciding to hold virtual classes. The absence of internet access and the inability to afford a laptop, among other devices, can lead to significant feelings of discrimination. This situation not only results in academic setbacks but can also cause psychological harm that may be irreparable.

Another factor cited as discrimination was differences in students' ability to use technology. In a study by Pete et al. (2017) conducted in a quasi-experimental study of first-year nursing students with virtual education, only 57% of participants said they had been trained to use a computer, and only 42% could send an email indicating that students were not proficient in using e-learning. Finally, this study showed that participants are not sufficiently prepared to use technology to meet the requirements of e-learning. Therefore, assessing the readiness of using electronic tools before accepting and implementing e-learning is of great importance for the success of educational programs (Pete et al., 2017). In this regard, in a study, 53.4% of students cited a lack of information retrieval skills as their main challenge, and finally, 50.4% of students cited a lack of necessary skills in using the internet as their main challenge. In addition, the percentage of respondents living in rural areas who face challenges in using information and communication technology was more than urban respondents (Bello et al., 2017). Also, in a study aimed at discovering the barriers to e-learning at Hormozgan University of Medical Sciences in Iran, the level of awareness of the e-learning programs among student groups was only 43% (Nakhaee & Nasiri, 2017). Consequently, it is recommended that university education officials incorporate additional units into their official educational curriculum to better train nursing students in the use of electronic tools.

In this study, teachers "skills in using technology and disregard for students" criticisms intensified their sense of discrimination. According to a study by Harerimana and Mtshali (2018), the success of e-learning in nursing depends on the readiness, knowledge, and skills of professors in educational design and use of technology (Harerimana & Mtshali, 2018), while in a study 66.7% of professors stated have not received e-learning programs in the past (Lakbala, 2016). Also, in the study of Keshavarzi et al. (2019), the participants believed that the most important problem for professors is that they are accustomed to the old system, and it is difficult for them to work with computers and advanced systems. They are unfamiliar and do not like to tolerate this change due to energy expenditure, so they resist it, so the university should supervise these professors (Keshavarzi et al., 2019). In this study, problems related to the internet, lack of access to library resources, and appropriate e-learning infrastructure were identified as other factors causing discrimination among students. In a review study conducted by Sadeghi Mahali et al. (2023) to compare the challenges of e-learning before and after COVID-19, an important part of the challenges related to e-learning was the lack of access to magazines and e-books, insufficient equipment for recording and holding video conferences, the existence of unsuitable audio spaces and acoustic rooms required the provision of educational content, as well as

the limitation on the number of computers (Sadeghi Mahali et al., 2023). In the same study, which was conducted in Egypt to examine the challenges of e-learning, 84.3% of students complained about poor internet and 68.7% of students complained about access to required information (Bello et al., 2017). The main challenges, according to the participants, were the poor access to the internet, so that students had difficulty uploading files and also sometimes did not have access to exams. Therefore, they did not pass that test and could not pass the course successfully. And this problem was more for the villagers because the internet is much weaker in those areas (Ilonga et al., 2020). In the study of Salmani et al. (2022), students' educational files frequently encountered various problems such as low sound quality, extra sounds during sound recording, uncertain start and end of the sound, very short and concise slides, high volume slides, unfamiliar English terms on slides and high volume, and sometimes even loaded incorrectly. In this study, students' hardware problems included power outages, internet outages, lack of laptops, mobile phones without advanced technologies, and lack of access to reference books (Salmani et al., 2022). In another study, the findings showed that there was an insufficient infrastructure to admit a large number of regular students to e-learning, including inadequate classrooms, poor libraries with inadequate online textbooks, a lack of an ICT lab and skills lab, and a lack of computers for all students (Harerimana & Mtshali, 2018). Therefore, it is suggested that, prior to conducting virtual training, the proficiency of professors in using electronic tools be assessed, and training sessions be organized accordingly. Additionally, due to the absence of physical presence among students and their resulting limited access to educational resources, it is advisable for university officials to provide free internet access for students and facilitate the downloading of reliable materials from educational websites. Furthermore, professors should be required to upload educational resources to online platforms.

Implications for Nursing Education

The findings of this study have significant implications for nursing education, particularly in the context of e-learning. First, addressing the identified challenges such as inadequate interaction between students and professors, lack of feedback, and insufficient infrastructure can enhance the quality of virtual education and reduce feelings of discrimination among nursing students. Educational institutions should prioritize blended learning models that combine the strengths of face-to-face and virtual education to ensure more inclusive and equitable learning experiences.

Second, nursing faculties need to receive targeted training in the design and implementation of e-learning to develop technological proficiency and pedagogical skills. Providing ongoing professional development programs can help professors better engage with students and address their concerns effectively.

Third, institutions should invest in improving e-learning infrastructure, including high-speed internet, access to digital resources, and advanced technological tools, especially for students in rural or economically disadvantaged areas. Ensuring equitable access to resources can mitigate disparities and promote a more inclusive educational environment.

Lastly, nursing education policymakers must develop guidelines to assess students' readiness for e-learning and provide necessary support, such as training in digital literacy and access to technological tools. These measures can empower students to adapt to virtual learning environments and foster a sense of belonging and fairness.

By addressing these implications, nursing educators and administrators can create a more supportive and equitable e-learning environment, ultimately enhancing the learning outcomes and satisfaction of nursing students.

Future Lines of Research

This study has highlighted key gaps in understanding discrimination in e-learning, which can be addressed in future research. Future studies could focus on exploring specific strategies or interventions that may reduce perceived discrimination in virtual education. Additionally, research could examine the impact of discrimination on long-term academic and professional outcomes, such as student motivation, self-confidence, and career progression. Comparative studies across different cultural and institutional contexts could help identify universal and context-specific factors contributing to discrimination in e-learning. Further quantitative and mixed-methods research is also needed to provide more generalizable insights into the issues identified in this study

Limitations

One of the limitations of this study—like all qualitative studies—was about generalizing the result of this study. As such, it may not be representative of the experiences of all the nursing profession members in Iran or the world. However, it provides very profound insights into discrimination in virtual learning that are often overlooked. In addition, it is highlighted that this study is based on qualitative data and thus the aim was in-depth understanding rather than reaching statistical generalization. Limitations of our study proposed the need for conducting further studies with larger and mixed groups and in different cultures

Conclusion

Our study showed that even virtual and distance education can cause discrimination among students. And this becomes even more important after the COVID-19 pandemic and the spread of e-learning around the world. Discrimination in many cases causes a lack of interest and motivation or reduces students' self-confidence. This can negatively

affect the training of efficient and professional people (nurses) for the future and cause problems for the health care system.

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Statements and Declarations

Ethics Approval and Consent to Participate

This research was proposed at the Study Center of Isfahan University's Nursing Faculty and received approval from the ethics committee with code IR.MUI.NUREMA.REC.1401.087. In this study, both oral and written consent were obtained from each participant after explaining the study's purpose. All participants were assured of the complete confidentiality of their interviews, and recordings of the interviews were scheduled for destruction after transcription and analysis. Additionally, nursing students were informed that their comments regarding the use or non-use of the nursing process would not impact their grades.

Author Contributions

Z.H.J.: conducting interviews with participants, conceptualization, analyzing data, methodology, writing, reviewing, and editing the article, formal analysis. A.Sh.: conducting interviews with participants, writing interviews and manuscript, analyzing demographic data, and completing article. K.Gh.: reviewing and editing in terms of clinical aspects. S.F.: conceptualization and formal analysis. S.B.: supervision.

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Supplemental Material

Supplemental material for this article is available online.

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